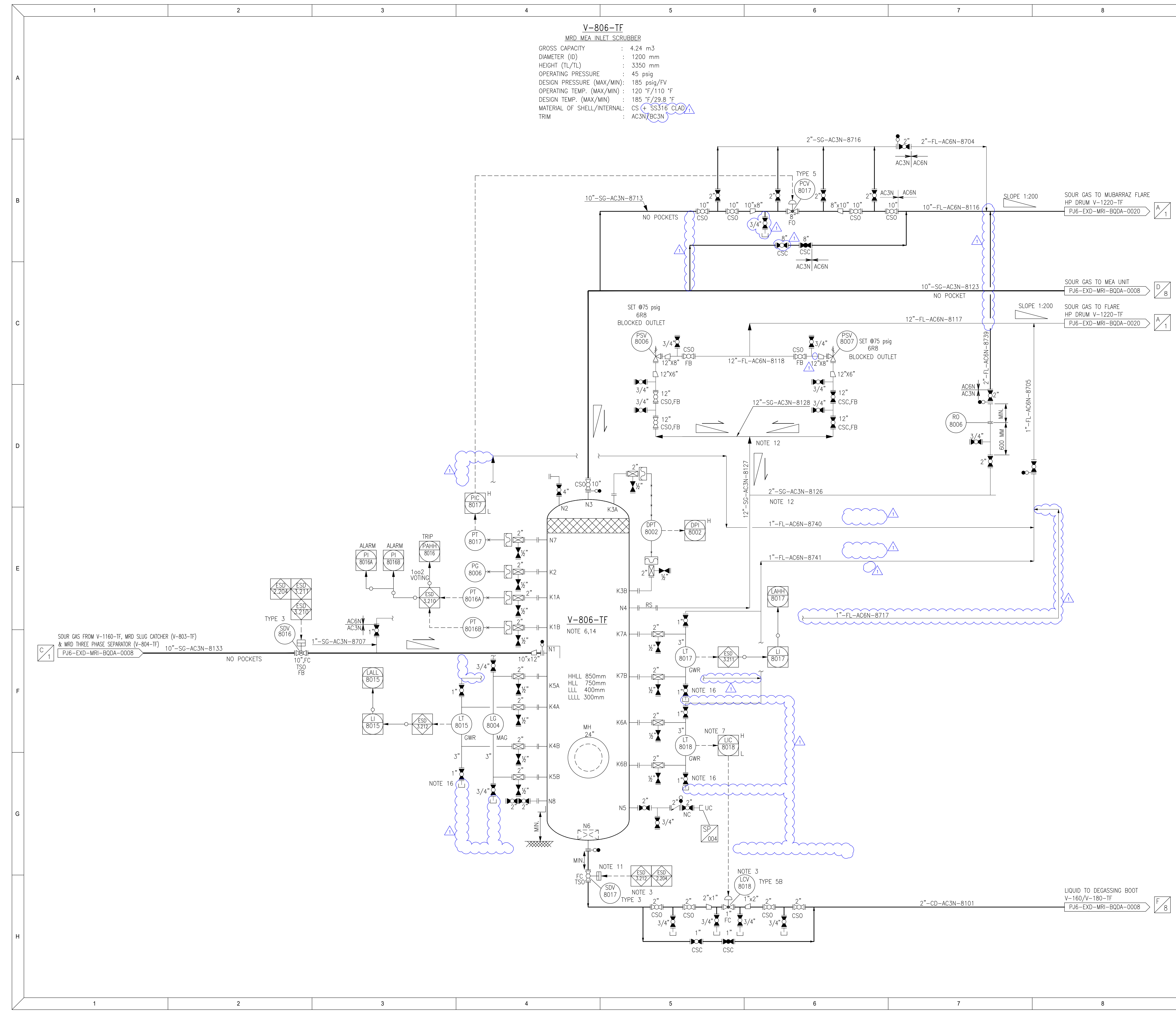




NOTES

- ALL INSTRUMENT TAG NUMBERS SHALL BE SUFFIXED BY SITE SYMBOL CODE "TF" FOR MUBARRAZ ISLAND.
- DELETED.
- SDV-8017 AND LCV-8018 TO BE LOCATED AT MINIMUM DISTANCE FROM V-806-TF.
- ALL DRAINS SHALL BE CONNECTED TO CLOSED DRAIN SYSTEM AND ALL VENTS SHALL BE CONNECTED TO FLARE.
- FUEL GAS SYSTEM IS DESIGNED FOR COMPLETE GAS FROM MUBARRAZ ISLAND.
- DEMISTER PAD SHALL BE DESIGN TO REMOVE 99.5% OF DROPLETS ABOVE 10 MICRONS.
- MODULATING LEVEL CONTROL IS CONSIDERED.
- DELETED.
- DELETED.
- DELETED.
- ALL THE VALVES AROUND THE VESSEL, SHALL BE CLOSE TO THE VESSEL. DEAD LEG SHALL BE AVOIDED IN THE DRAIN LINES.
- SYMMETRICAL PIPING ARRANGEMENT SHALL BE PROVIDED FOR THE PSVs.
- MINIMUM RATING OF INSTRUMENTS SHALL BE 300#. FOR NOZZLE RATING REFER EQUIPMENT DATASHEET.
- EQUIPMENT IS FREE ISSUE ITEM BY ADOC AND ASSOCIATED INSTRUMENTS ARE IN SCOPE OF DETAILED ENGINEERING CONTRACTOR.
- DELETED.
- LG AND LT INSTRUMENT ISOLATION DBB, VENT AND DRAIN VALVE SCOPE DETAILS REFER INSTRUMENT HOOKUP DRAWING: PJ6-EXD-GEN-YZ0A-0002

LEGEND

 SCOPE CLOUD

 REVISION CLOUD

REFERENCE DOCUMENTS / DRAWINGS

S.No.	DOCUMENT / DRAWING TITLE	DOC / DWG No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002
2	PFD MUBARRAZ ISLAND (1/2)	PJ6-EXD-MRI-BQDA-0005

THIS DRAWING HAS BEEN DEVELOPED FROM FEED DRAWING NO. PJ6-DF-MRI-BQDA-0001, REV 1. CLOUD PORTION DENOTES SCOPE OF WORK UNDER DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT "PJ-2025-CR-002"

1	30/01/2026	RE-APPROVED FOR CONSTRUCTION	DVS	MCS	PKR
0	28/11/2025	APPROVED FOR CONSTRUCTION	TIM	MCS	PKR
C	12/09/2025	ISSUED FOR DESIGN	TNM	ASL	PKR
B	29/07/2025	ISSUED FOR HAZOP	ZAS	ASL	PKR
A	03/07/2025	ISSUED FOR REVIEW	ZAS	ASL	PKR

REV.	DATE	DESCRIPTION	DR'N	CH'D	AP'D
SCALE: NTS LOCATION: MUBARRAZ ISLAND COMPANY PROJECT NO. PJ-2025-CR-002					

ENGINEERING CONSULTANT:



REJLERS
REJLERS TOWER 13TH FLOOR
HAMDAN STREET, PO BOX 39317
ABU DHABI, UAE

ENG CONSULTANT PROJECT NO: 5901055

PMC : DORSCH HOLDING GmbH - ABU DHABI

COMPANY:



ABU DHABI OIL CO., LTD.

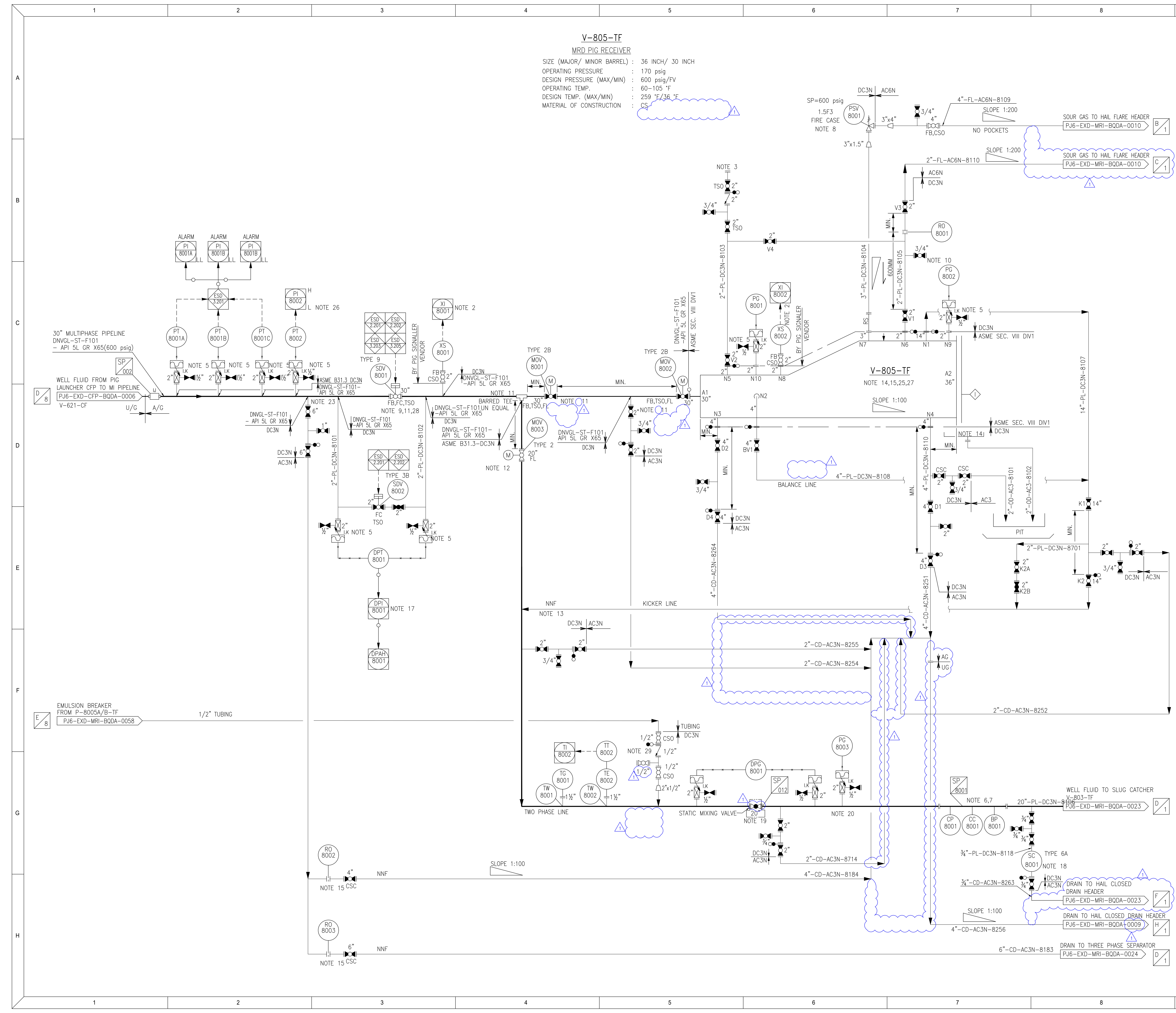
PROJECT:
DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)

DRG. TITLE:
PIPING AND INSTRUMENTATION DIAGRAM
NEW MEA INLET SCRUBBER

DWG. NO.	PJ6-EXD-MRI-BQDA-0021	REV NO.	1	SHEET NO.	001
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Filename : PJ6-EXD-MRI-BQDA-0021_1.DWG

SHT. SIZE: A1



NOTES

1. ALL INSTRUMENT TAG NUMBERS ARE SUFFIXED WITH "TF" FOR MUBARRAZ ISLAND.

2. INTRUSIVE TYPE PIG SIGNALLER.

3. NITROGEN PURGING CONNECTION/WATER FLUSHING CONNECTION.

4. QUICK OPENING END CLOSURE DOOR WILL BE EQUIPPED WITH A PRESSURE LOCKING DEVICE TO PREVENT OPENING OF DOORS WHEN THE PIG TRAP IS PRESSURIZED. A PRESSURE WARNING SAFETY INTERLOCK DEVICE SHALL ALSO BE PROVIDED AS PER AGES-SP-06-008.

5. INSULATING KIT IS PROVIDED FOR DISSIMILAR METAL JOINTS.

6. CORROSION COUPON & CORROSION PROBE WILL BE PROVIDED ON REMOVABLE SPOOL.

7. CORROSION COUPON, CORROSION PROBE & BIO PROBE WILL BE RETRACTABLE UNDER PRESSURE. THE ACCESS FITTINGS ARE LOCATED AT THE HORIZONTAL LINE AT 6 O' CLOCK POSITION.

8. WAREHOUSE SPARE PSV (TAG PSV-8008-TF) TO BE PROVIDED.

9. CLOSING OF SDV (SDV-8001) WILL GENERATE AN ALARM IN DCS.

10. PRESSURE INDICATOR (PG-8002) IS VISIBLE FROM PIG LAUNCHER DOOR. SCALE OF PRESSURE INDICATOR SHALL BE FROM 0 TO 15 psig, BUT PRESSURE RATED TO LINE PRESSURE.

11. INTERNAL DIAMETER TO BE IDENTICAL TO PIPELINE INTERNAL DIAMETER.

12. VALVE TO BE PROVIDED WITH INCHING FACILITY AND MECHANICAL STOP.

13. AVOID DEAD LEG.

14. VOLUME OF DRIP TRAY SHALL BE MINIMUM 5% OF BARREL VOLUME.

15. LINE PROVIDED TO RECOVER OIL FROM PIPELINE IN CASE OF EMERGENCY.

16. ALL DRAINS SHALL BE CONNECTED TO CLOSED DRAIN SYSTEM AND ALL VENTS SHALL BE CONNECTED TO FLARE.

17. HIGH DIFFERENTIAL PRESSURE OF LESS THAN 100 PSI SHALL GIVE PERMISSIVE TO OPEN SDV-8001.

18. SAMPLE COLLECTION POINT SHALL BE ACCESSIBLE AT GROUND LEVEL.

19. STATIC MIXING VALVE SHALL BE SUPPLIED BY PIG TRAP VENDOR.

20. PRESSURE GAUGE SHALL BE READABLE FROM MIXING VALVE.

21. MINIMUM RATING OF INSTRUMENTS SHALL BE 300# FOR NOZZLE RATING.

22. DELETED.

23. SMALL BORE PIPING BRANCH UNDER ASME B 31.3 CODE WELDED TO DNV-ST-F101 PIPELINE WILL BE HYDROSTATICALLY TESTED JOINTLY WITH THE PIPELINE ACCORDING TO DNV CODE REQUIREMENT UP TO FIRST FLANGED JOINT AVAILABLE.

24. EQUIPMENT IS FREE ISSUE ITEM BY ADOC AND ASSOCIATED INSTRUMENTS ARE IN SCOPE OF DETAILED ENGINEERING CONTRACTOR.

25. PIPING TO PROVIDE ENOUGH SPACE AT THE END CLOSURE DOOR TO INSERT AND COLLECT PIG AT PIG LAUNCHER AND RECEIVER.

26. PI-8002L IS SAFETY CRITICAL ALARM.

27. PIG TRAPS SHALL BE PROVIDED WITH A PERMANENT JIB CRANE AND CHAIN PULLY BLOCK AT SITE.

28. CONSIDER 1A VOLUME BOTTLE FOR SDV-8001.

29. TWO DISSIMILAR TYPE CHECK VALVE SHALL BE USED AT INJECTION PUMP DISCHARGE AND AT THE INJECTION POINT.

LEGEND

SCOPE CLOUD

REVISION CLOUD

REFERENCE DOCUMENTS / DRAWINGS

S.No.	DOCUMENT / DRAWING TITLE	DOC / DWG No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002
2	PFD MUBARRAZ ISLAND (1/2)	PJ6-EXD-MRI-BQDA-0005

THIS DRAWING HAS BEEN DEVELOPED FROM FEED DRAWING NO. PJ6-DF-MRI-BQDA-0002 REV 1. CLOUD PORTION DENOTES SCOPE OF WORK UNDER DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT "PJ-2025-CR-002"

30/01/2026

RE-APPROVED FOR CONSTRUCTION

DVS

MCS

PKR

05/12/2025

APPROVED FOR CONSTRUCTION

TIM

MCS

PKR

26/09/2025

ISSUED FOR DESIGN

TNM

ASL

PKR

29/07/2025

ISSUED FOR HAZOP

ZAS

ASL

PKR

03/07/2025

ISSUED FOR REVIEW

ZAS

ASL

PKR

REV.

DATE

DESCRIPTION

DR'N

CH'D

AP'D

SCALE: NTS

LOCATION: MUBARRAZ ISLAND

COMPANY PROJECT NO. PJ-2025-CR-002

ENGINEERING CONSULTANT:

REJLERS

REJLERS TOWER 13TH FLOOR
HAMDAN STREET, PO BOX 39317
ABU DHABI, UAE

ENG CONSULTANT PROJECT NO: 5901055

PMC : DORSCH HOLDING GmbH - ABU DHABI

COMPANY:

ABU DHABI OIL CO., LTD.

PROJECT:

DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)

DRG. TITLE:

PIPING AND INSTRUMENTATION DIAGRAM
PIG RECEIVER FOR CFP TO MI PIPELINE

DWG. NO.

PJ6-EXD-MRI-BQDA-0022

REV NO. 1

SHEET NO. 001

Filename : PJ6-EXD-MRI-BQDA-0022.1.DWG

12

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V-804-TF

MRD THREE PHASE SEPARATOR

NOMINAL CAPACITY : 120 m³
LENGTH (T/T) : 12000 mm
DIAMETER (ID) : 3400 mm
OPERATING PRESS. : 110 psig
DESIGN PRESS. (MAX/MIN) : 185 psig/ FV
OPERATING TEMP.(MAX/MIN) : 105/60 °F
DESIGN TEMP. (MAX/MIN) : 185 °F/25 °F
MATERIAL OF SHELL/INTERNAL: CS + LINING
TRIM : AC3N

SOUR GAS FROM MRD OIL
SLUG CATCHER V-803-TF
PJ6-DF-MRI-BQDA-0003

6"-SG-AC3N-8112

10"-SG-AC3N-8122

SOUR GAS TO MRD MEA
INLET SCRUBBER V-806-TF
PJ6-EXD-MRI-BQDA-0014

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

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PJ6-EXD-MRI-BQDA-0010

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PJ6-EXD-MRI-BQDA-0010

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PJ6-EXD-MRI-BQDA-0010

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PJ6-EXD-MRI-BQDA-0010

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PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

NOTES

- DELETED.
- SAMPLE COLLECTION POINT SHALL BE ACCESSIBLE AT GROUND LEVEL.
- NO POCKET. MINIMUM BEND.
- HC LIQUID & WATER OUTLET NOZZLES TO BE PROVIDED WITH VORTEX BREAKER.
- INTERFACE LOW LEVEL TRIP TO AVOID OIL GAS BLOW-BY CASE.
- SEPARATOR INTERNALS: HALF OPEN PIPE (INLET DEVICE) SUBMERGED WEIR, MIST ELIMINATOR, VORTEX BREAKERS AND PERFORATED PLATE.
- LI-8005 READABLE FROM LCV-8005 BYPASS GLOBE VALVE.
- MAGNETIC TYPE FLOWMETER.
- DRAIN POINT TO FACILITATE REMOVAL OF SLUDGE/SAND DURING MAINTENANCE.
- CORROSION COUPON & CORROSION PROBE WILL BE PROVIDED ON REMOVABLE SPOOL.
- CORROSION COUPON AND CORROSION PROBE WILL BE RETRACTABLE UNDER PRESSURE. THE ACCESS FITTINGS ARE LOCATED AT THE HORIZONTAL LINE AT 6 O' CLOCK POSITION.
- ULTRASONIC FLOWMETER.
- ALL THE VALVES AROUND THE VESSEL, SHALL BE CLOSE TO THE VESSEL. DEAD LEG SHALL BE AVOIDED IN THE DRAIN LINES.
- GASKET FOR ALL THE VALVES ASSOCIATED WITH THE VESSEL SHALL BE SUITABLE FOR STEAM.
- SYMMETRICAL PIPING ARRANGEMENT SHALL BE CONSIDERED FOR THE PSVS.
- LG AND LT INSTRUMENT ISOLATION DBB, VENT AND DRAIN VALVE SCOPE DETAILS REFER INSTRUMENT HOOKUP DRAWING.PJ6-EXD-GEN-Y20A-0002
- PROVISION FOR JETTING WATER CONNECTION.
- MINIMUM STRAIGHT LENGTH REQUIRED AT UPSTREAM AND DOWNSTREAM TO BE 10D AND 5D RESPECTIVELY.
- ALL DRAINS SHALL BE CONNECTED TO CLOSED DRAIN SYSTEM AND ALL VENTS SHALL BE CONNECTED TO FLARE.
- MESH PAD TYPE DEMISTER SHALL BE PROVIDED AT THE GAS OUTLET.
- LI-8008 READABLE FROM LCV-8008 BYPASS GLOBE VALVE.
- MINIMUM RATING OF INSTRUMENTS SHALL BE 300#. FOR NOZZLE RATING REFER EQUIPMENT DATASHEET.
- NOZZLE SHALL BE KEPT AT THE CENTER OF THE VESSEL.
- EQUIPMENT IS FREE ISSUE ITEM BY ADCC AND ASSOCIATED INSTRUMENTS ARE IN SCOPE OF DETAILED ENGINEERING CONTRACTOR.
- FLOW TO MRD PRODUCED WATER TREATMENT PACKAGE SHALL BE LIMITED TO 81,440 BWPD.
- DELETED.
- 5 NO OF SAMPLE THIEF PROVISION TO BE PROVIDED IN THE MRD THREE PHASE SEPERATOR.
- FI-8003H IS SAFETY CRITICAL ALARM.

LEGEND

- SCOPE CLOUD
- REVISION CLOUD

REFERENCE DOCUMENTS / DRAWINGS

S.No.	DOCUMENT / DRAWING TITLE	DOC / DWG No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002
2	PFD MUBARRAZ ISLAND (1/2)	PJ6-EXD-MRI-BQDA-0005

THIS DRAWING HAS BEEN DEVELOPED FROM FEED DRAWING NO. PJ6-DF-MRI-BQDA-0004 REV 1. CLOUD PORTION DENOTES SCOPE OF WORK UNDER DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT "PJ-2025-CR-002"

REV.	DATE	DESCRIPTION	DR'N	CH'D	AP'D
1	30/01/2026	RE-APPROVED FOR CONSTRUCTION	DVS	MCS	PKR
0	28/11/2025	APPROVED FOR CONSTRUCTION	TIM	MCS	PKR
C	12/09/2025	ISSUED FOR DESIGN	TNM	ASL	PKR
B	29/07/2025	ISSUED FOR HAZOP	SRK	ASL	PKR
A	03/07/2025	ISSUED FOR REVIEW	SRK	ASL	PKR
SCALE: NTS			LOCATION: MUBARRAZ ISLAND		
ENGINEERING CONSULTANT:			COMPANY PROJECT NO. PJ-2025-CR-002		

REJLERS
REJLERS TOWER 13TH FLOOR
HAMDAN STREET,PO BOX 39317
ABU DHABI, UAE

ENG CONSULTANT PROJECT NO:5901055

PMC : DORSCH HOLDING GmbH - ABU DHABI

COMPANY:

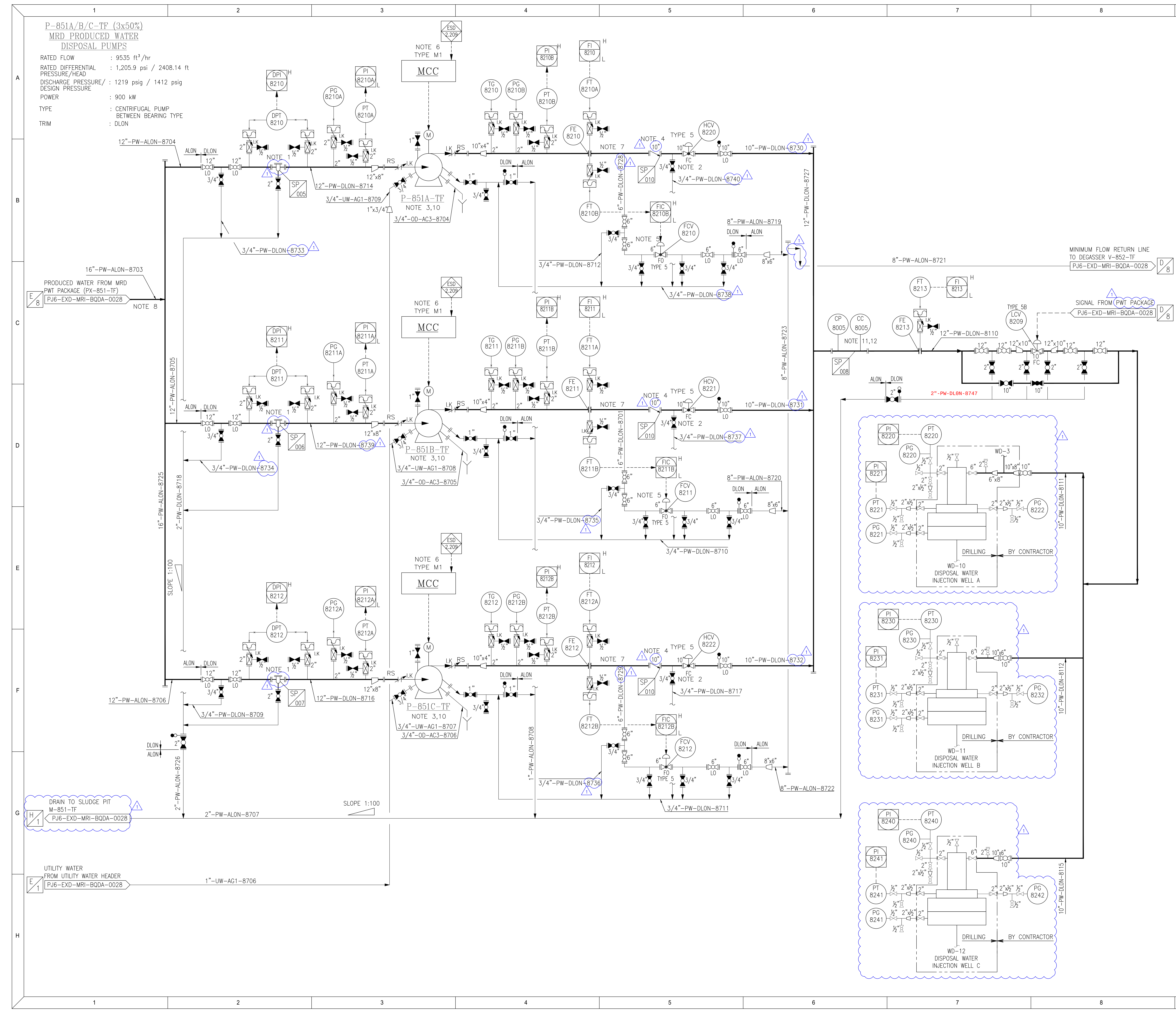
ABU DHABI OIL CO.,LTD.

PROJECT:
DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)

DRG. TITLE:
PIPING AND INSTRUMENTATION DIAGRAM
NEW MUBARRAZ THREE PHASE SEPARATOR

DWG. NO. PJ6-EXD-MRI-BQDA-0024

Filename : PJ6-EXD-MRI-BQDA-0024_1.DWG



NOTES

1. PERMANENT STRAINER (BUCKET TYPE).

2. VALVE TO OPEN AND CLOSE SLOWLY TO PREVENT HAMMER EFFECT. VALVE TO OPEN WITH ACTIVATION OF ASSOCIATED PUMP.

3. PUMP SUPPLIED WITH API 610 PLAN 53B MECHANICAL SEAL.

4. NON SLAM TYPE CHECK VALVE (WITH 5D UPSTREAM & 3D DOWNSTREAM).

5. HIGH INTEGRITY ANTI CAVITY TRIM.

6. PUMP CONFIGURATION: 3 X 50% (2 DUTY + 1 STANDBY). STANDBY PUMP SHALL START AUTOMATICALLY IN THE EVENT OF FAILURE OF EITHER DUTY PUMP.

7. ORIFICE TAPING TYPE SHALL BE FLANGE TAPS. THE TAPPINGS ON ORIFICE FLANGES FOR MONITORING/CONTROL AND ESD/TRIP APPLICATIONS SHALL NOT BE ON THE SAME POINTS. FOR TAPPING ORIENTATION, REFER TO INSTRUMENT HOOK-UP DRAWING PJ6-EXD-GEN-Y20A-0002.

8. PUMP SUCTION AND DISCHARGE TO BE SYMMETRICAL ARRANGEMENT.

9. ALL PUMP CONTROL WILL BE FROM VENDOR PACKAGE PLC.

10. PRODUCED WATER DISPOSAL PUMP AND RELATED INSTRUMENTS WILL BE SUPPLIED BY PUMP VENDOR AND ALL OTHER INSTRUMENT ARE UNDER PC CONTRACTOR.

11. CORROSION COUPON & CORROSION PROBE WILL BE PROVIDED ON REMOVABLE SPOOL.

12. CORROSION COUPON AND CORROSION PROBE WILL BE RETRACTABLE UNDER PRESSURE. THE ACCESS FITTINGS ARE LOCATED AT THE HORIZONTAL LINE AT 6 O' CLOCK POSITION.

LEGEND

REVISION CLOUD

REFERENCE DOCUMENTS / DRAWINGS

S.No.	DOCUMENT / DRAWING TITLE	DOC / DWG No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002
2	PFD MUBARRAZ ISLAND (2/2)	PJ6-EXD-MRI-BQDA-0006

1

30/01/2026

RE-APPROVED FOR CONSTRUCTION

DVS

MCS

PKR

0

16/12/2025

APPROVED FOR CONSTRUCTION

TIM

MCS

PKR

C

12/09/2025

ISSUED FOR DESIGN

TNM

ASL

PKR

B

29/07/2025

ISSUED FOR HAZOP

ZAS

ASL

PKR

A

03/07/2025

ISSUED FOR REVIEW

ZAS

ASL

PKR

REV.	DATE	DESCRIPTION	DR'N	CH'D	AP'D
SCALE: NTS	LOCATION: MUBARRAZ ISLAND	COMPANY PROJECT NO. PJ-2025-CR-002			

ENGINEERING CONSULTANT:

REJLERS

REJLERS TOWER 13TH FLOOR
HAMDAN STREET, PO BOX 39317
ABU DHABI, UAE

ENG CONSULTANT PROJECT NO: 5901055

PMC : DORSCH HOLDING GmbH - ABU DHABI

COMPANY:

ABU DHABI OIL CO., LTD.

PROJECT:

DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)

DRG. TITLE:

PIPING AND INSTRUMENTATION DIAGRAM
NEW PRODUCED WATER DISPOSAL
PUMPS (P-851 A/B/C-TF)

DWG. NO.	PJ6-EXD-MRI-BQDA-0042	REV NO.	SHEET NO.
		1	001

Filename : PJ6-EXD-MRI-BQDA-0042_1.DWG

SHT. SIZE: A1